

UNIVERSIDAD NACIONAL DE INGENIERÍA
FACULTAD DE CIENCIAS
ESCUELA PROFESIONAL DE FÍSICA



EL INVARIANTE $\mathbb{Z}_2$

TÓPICOS DE ESPECIALES IB

Estudiante:
César Octavio Lévano Apari

Profesor:
Prof. Dr. Antonio Zegarra

Octubre 2025

Agradecimientos

Resumen

...

Contents

Resumen	III
1 Introducción	1
2 Aislantes topológicos	3
2.1 Introducción a los haces vectoriales	3
2.2 Zona de Brillouin y toroide de Brillouin	3
2.2.1 Ejemplos	3
2.3 Simetrías discretas	4
2.3.1 Simetría de inversión temporal	4
2.3.2 Simetría de hueco-partícula	4
2.3.3 Simetría de paridad	4
3 Introduction to K-theory	6
3.1 Classification of vector bundles	6
3.2 Classification of matrices	6
3.3 The Grothendieck group	6
3.4 Equivalence between $\text{Vec}(X)$ and $\text{Idem}(\mathcal{C}(X))$	6
3.5 Quaternionic K-theory	6
3.6 Equivariant vector bundles	7
3.7 Equivariant K-theory	7
4 El modelo de Kane-Mele	9
4.1 El modelo de Haldane	9
4.2 El término de Rashba	9
4.3 Deducción del invariante $\mathbb{Z}_2$	9
5 Conclusiones y Perspectivas	11

Chapter 1

Introducción

Chapter 2

Aislantes topológicos

2.1 Introducción a los haces vectoriales

2.2 Zona de Brillouin y toroide de Brillouin

2.2.1 Ejemplos

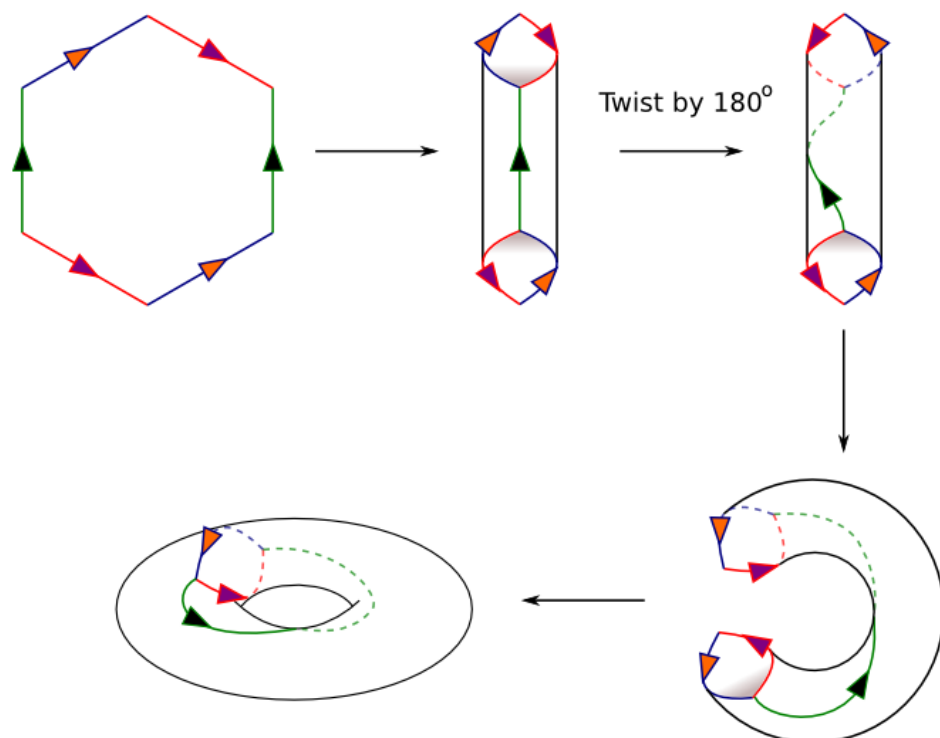


Figure 2.1: Gluing process of the hexagon. Extracted from [1]

2.3 Simetrías discretas

2.3.1 Simetría de inversión temporal

2.3.2 Simetría de hueco-partícula

2.3.3 Simetría de paridad

Chapter 3

Introduction to K-theory

3.1 Classification of vector bundles

3.2 Classification of matrices

3.3 The Grothendieck group

Definition 3.3.1 (Abelian monoid). *An abelian monoid is the pair $(A, +)$ such that:*

- $(a + b) + c = a + (b + c)$ for all $a, b, c \in A$;
- $a + b = b + a$, for all $a, b \in A$;
- $\exists 0 \in A$ such that for all $a \in A$, $0 + a = a$.

3.4 Equivalence between $\text{Vec}(X)$ and $\text{Idem}(\text{C}(X))$

Definition 3.4.1 ($K^0(X)$). *Let X be a compact Hausdorff. The Grothendieck group generated from $\text{Vec}(X)$ is denoted by $K^0(X)$.*

3.5 Quaternionic K-theory

Definition 3.5.1 (Quaternionic vector bundle). *A quaternionic vector bundle over (X, τ) is a complex vector bundle $\pi: \mathcal{E} \rightarrow X$ endowed with a homeomorphism $\Theta: \mathcal{E} \rightarrow \mathcal{E}$ such that:*

- the projection π is equivariant in the sense that $\pi \circ \Theta = \tau \circ \pi$;
- Θ is antilinear on each fiber, i.e. $\Theta(\lambda p) = \bar{\lambda} \Theta(p)$ for all $\lambda \in \mathbb{C}$ and $p \in \mathcal{E}$;
- Θ^2 acts fiberwise as multiplication by -1 , namely $\Theta^2|_{\mathcal{E}_x} = -\mathbb{1}_{\mathcal{E}_x}$.

3.6 Equivariant vector bundles

Definition 3.6.1 (G-space). *Let G be a finite group. An action of G on a topological space X is a continuous mapping $G \times X \rightarrow X$ given by*

$$(g, x) \mapsto g \cdot x \quad (3.1)$$

with the properties

- $1 \cdot x = x$ for all $x \in X$;
- $(gh) \cdot x = g \cdot (h \cdot x)$ for all $g, h \in G$ and all $x \in X$.

A pair consisting of a space X and an action G on it is called a G-space.

3.7 Equivariant K-theory

Definition 3.7.1 (G-vector bundle). *Let G be a compact Lie group and X be a G -manifold. A G -vector bundle is a G -map $\xi: E \rightarrow X$ such that:*

- ξ has the structure of a vector bundle;
- the map $g: E_x \rightarrow E_{gx}$ is linear for all $g \in G$ and $x \in X$.

Definition 3.7.2 (Equivariant K-theory). *The equivariant K-theory of a G -space X is defined as:*

$$K_G(X) \equiv K(\text{Vec}_G(X)). \quad (3.2)$$

where $\text{Vec}_G(X)$ is the collection of isomorphism classes of G -vector bundles over X .

Chapter 4

El modelo de Kane-Mele

4.1 El modelo de Haldane

4.2 El término de Rashba

$$H_R = \int \Psi^\dagger(\mathbf{r}) \gamma (p_x \sigma_y - p_y \sigma_x) \Psi(\mathbf{r}) d^3r \quad (4.1)$$

4.3 Deducción del invariante $\mathbb{Z}_2$

Chapter 5

Conclusiones y Perspectivas

Bibliography

- [1] C. Victor, *Flat life*, Cornell University Mathematics Department, 2009. [Online]. Available: <https://pi.math.cornell.edu/mec/Winter2009/Victor/part1.htm>